

Lecture: 15 Expansion of Functions

By Dr. A.K. Misra

Some Expansions

1. $\sin x$: For all $n \in \mathbb{N}$, $f^n(x) = \sin\left(\frac{n\pi}{2} + x\right)$ exists for all $x \in \mathbb{R}$.

By Taylor's theorem with Lagrange's form of remainder after n terms, for any non-zero $x \in \mathbb{R}$,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{n-1}(0) + R_n(x),$$

where $R_n(x) = \frac{x^n}{n!}f^n(\theta x)$ for some real number θ satisfying $0 < \theta < 1$.

$$\begin{aligned} f(0) = 0, f^n(0) &= \sin \frac{n\pi}{2} = 0 \text{ if } n \text{ is even, } 1 \text{ if } n = 4k + 1, k \text{ being an integer,} \\ &= -1 \text{ if } n = 4k + 3, k \text{ being an integer.} \end{aligned}$$

Therefore,

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{x^n}{n!} \sin\left(\frac{n\pi}{2} + \theta x\right).$$

The infinite series will converge to $f(x)$ for those non-zero real x for which

$$\lim_{n \rightarrow \infty} R_n = 0.$$

Now, $|R_n| = \left| \frac{x^n}{n!} \sin\left(\frac{n\pi}{2} + \theta x\right) \right| \leq \frac{|x|^n}{n!}$.

Let $u_n = \frac{x^n}{n!}, x \neq 0$. Then $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0$. This proves that $\lim_{n \rightarrow \infty} u_n = 0$.

Hence, $\lim_{n \rightarrow \infty} |R_n| = 0$ for all real $x \neq 0$ and this implies $\lim_{n \rightarrow \infty} R_n = 0$ for all real $x \neq 0$.

At $x = 0$, the convergence holds trivially.

Therefore,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \text{ for all real } x.$$

2. Let $f(x) = (1+x)^m, x \in \mathbb{R}$.

Case 1. Let m is a positive integer, then

$$\begin{aligned} f^n(x) &= m(m-1)(m-2)\dots(m-n+1)(1+x)^{m-n}, \text{ if } 1 \leq n < m. \\ &= m!, \text{ if } n = m \end{aligned}$$

$$= 0, \text{ if } n > m.$$

By Taylor's theorem with remainder after $m + 1$ terms, for any non zero $x \in \mathbb{R}$,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^m}{m!}f^{(m)}(0).$$

But $f(0) = 1$, $f^{(n)}(0) = \binom{m}{n}$ for $1 \leq n \leq m$. Therefore,

$$(1+x)^m = 1 + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m \text{ for all non-zero } x \in \mathbb{R}.$$

At $x = 0$, the equality holds trivially.

Therefore, $(1+x)^m = 1 + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m$ for all $x \in \mathbb{R}$

Thus, we obtain a finite series expansion in this case.

Case 2. Let m be not a positive integer.

In this case f is defined for all $x \neq -1$, if m be a negative integer and f is defined for all $x > -1$, if m be not an integer.

Considering all cases, f is defined for all $x > -1$; and for all $n \in \mathbb{N}$,

$$f^{(n)}(x) = m(m-1)\dots(m-n+1)(1+x)^{m-n} \text{ for all } x > -1.$$

By Taylor's theorem with Cauchy's form of remainder after n terms for any real non-zero $x > -1$,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + R_n(x),$$

where $R_n(x) = \frac{x^n(1-\theta)^{n-1}}{(n-1)!}f^{(n)}(\theta x)$, for some real θ satisfying $0 < \theta < 1$.

But $f(0) = 1$, $f^{(n)}(0) = m(m-1)\dots(m-n+1)$, $f^{(n)}(\theta x) = m(m-1)\dots(m-n+1)(1+\theta x)^{m-n}$.

Therefore, $f(x) = 1 + mx + \frac{m(m-1)}{2!}x^2 + \dots + \frac{m(m-1)\dots(m-n+1)}{(n-1)!}x^{n-1} + R_n(x)$, where $R_n(x) = \frac{x^n(1-\theta)^{n-1}}{(n-1)!}m(m-1)\dots(m-n+1)(1+\theta x)^{m-n}$

Since, for all $n \in \mathbb{N}$, $f^{(n)}(x)$ exists for all $x > -1$, the right hand side polynomial in takes the form of an infinite series as $n \rightarrow \infty$. The infinite series will converge to $f(x)$ for those non-zero $x > -1$ for which

$$\lim_{n \rightarrow \infty} R_n = 0.$$

$$\begin{aligned} R_n &= \frac{x^n(1-\theta)^{n-1}}{(n-1)!}m(m-1)\dots(m-n+1)(1+\theta x)^{m-n} \\ &= \frac{m(m-1)\dots(m-n+1).x^n}{(n-1)!} \left(\frac{1-\theta}{1+\theta x} \right)^{n-1} (1+\theta x)^{m-1}. \end{aligned}$$

Let $0 < |x| < 1$.

Let

$$u_n = \left| \frac{m(m-1)\dots(m-n+1)}{(n-1)!} \right| |x|^n.$$

Then, $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = |x| < 1$. Therefore, $\lim_{n \rightarrow \infty} u_n = 0$.

If $0 < |x| < 1$, then $0 < 1 - \theta < 1 + \theta x < 1 + \theta$.

Therefore, $0 < \frac{1-\theta}{1+\theta x} < 1$ and hence $0 < \left(\frac{1-\theta}{1+\theta x}\right)^{n-1} < 1$.

For all real x , $-|x| \leq x \leq |x|$. Hence, $-|x| < -\theta|x| \leq \theta x \leq \theta|x| < |x|$, since $0 < \theta < 1$.

Therefore, for $0 < |x| < 1$, $0 < 1 - |x| < 1 + \theta x < 1 + |x|$ and therefore

$$(1 + \theta x)^{m-1} < (1 + |x|)^{m-1} < 2^{m-1} \text{ if } m > 1$$

$$\text{and } (1 + \theta x)^{m-1} < (1 - |x|)^{m-1} \text{ if } m < 1.$$

Thus, for all $m (\neq 1)$, $(1 + \theta x)^{m-1}$ is bounded.

From above it follows that

$$\lim_{n \rightarrow \infty} |R_n| = 0,$$

when $0 < |x| < 1$.

The series

$$1 + mx + \frac{m(m-1)}{2!}x^2 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!}x^n + \dots$$

converges to $(1+x)^m$ for all non zero $x \in (-1, 1)$. At $x = 0$, the convergence holds trivially.

So,

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!}x^n + \dots$$

for all $x \in (-1, 1)$.

Definite integral as the limit of a sum

It is possible to regard the definite integral as the limit of the sum of a finite series of numbers, when the number of terms of the series tends to infinity while the terms of the series tends to zero. Thus we can define

$$\int_a^b f(x)dx$$

by the equation

$$\int_a^b f(x) = \lim_{n \rightarrow \infty} h [f(a) + f(a+h) + f(a+2h) + \dots + f\{a + (n-1)h\}],$$

where, $h = \frac{(b-a)}{n}$.

Summation of series

The value of the above integral can be written as,

$$\lim_{n \rightarrow \infty} \sum_{r=0}^{r=n-1} [f(a + rh)]h = \int_a^b f(x)dx, \text{ where } nh = b - a$$

or, more conveniently, by the formula derived from the above by putting $a = 0$ and $b = 1$, we have

$$\lim_{n \rightarrow \infty} \sum_{r=0}^{r=n-1} \left\{ f\left(\frac{r}{n}\right) \right\} \frac{1}{n} = \int_0^1 f(x)dx.$$

To write down the corresponding definite integral, replace $\frac{r}{n}$ by x , its common difference $\frac{1}{n}$ by dx , and $\lim_{n \rightarrow \infty} \sum$ by $\int$ and insert the value of $\frac{r}{n}$ for first and last terms as the lower and upper limits respectively in the integral. This procedure will evidently give us $\int f(x)dx$ taken between the proper limits.

Example. Find the limit, when $n \rightarrow \infty$, of the series

$$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n}.$$

Solution.

$$\begin{aligned} \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} &= \sum_{r=0}^{r=n} \frac{1}{n+r} \\ &= \sum_{r=0}^{r=n} \frac{\frac{1}{n}}{1 + \frac{r}{n}} \\ \Rightarrow \lim_{n \rightarrow \infty} \sum_{r=0}^{r=n} \frac{\frac{1}{n}}{1 + \frac{r}{n}} &= \int_0^1 \frac{dx}{1+x} = \log(1+x)|_0^1 = \log 2. \end{aligned}$$

Therefore,

$$\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right] = \log 2.$$

Example. Find the limit, as n tends to infinity, of the product

$$\left\{ \left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right\}^{1/n}.$$

Solution. Let

$$\begin{aligned} P &= \lim_{n \rightarrow \infty} \left\{ \left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right\}^{1/n}, \\ \Rightarrow \log P &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \left(1 + \frac{1}{n}\right) + \log \left(1 + \frac{2}{n}\right) + \log \left(1 + \frac{3}{n}\right) + \dots + \log \left(1 + \frac{n}{n}\right) \right], \end{aligned}$$

$$\begin{aligned}
\log P &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{r=n} \log \left(1 + \frac{r}{n} \right), \\
&\Rightarrow \log P = \int_0^1 \log(1+x) dx, \\
&\Rightarrow \log P = \log(1+x) \cdot x \Big|_0^1 - \int_0^1 \frac{x}{x+1} dx, \\
&= \log 2 - \int_0^1 \frac{1+x-1}{1+x} dx, \\
&= \log 2 - \int_0^1 1 \cdot dx + \int_0^1 \frac{1}{1+x} dx, \\
&= 2 \log 2 - \log e = \log \left(\frac{4}{e} \right), \\
&\Rightarrow \log P = \log \left(\frac{4}{e} \right) \Rightarrow P = \left(\frac{4}{e} \right).
\end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} \left\{ \left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right) \left(1 + \frac{3}{n} \right) \dots \left(1 + \frac{n}{n} \right) \right\}^{1/n} = \frac{4}{e}.$$

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.